04-21 第1篇

阅读理解（说明文） How Honeybees Talk with Their Dances

① On a warm spring afternoon, a single **honeybee** leaves the hive, flies over the garden, and finds a field of bright yellow flowers. Fifteen minutes later, dozens of bees from the same hive arrive at the same field. How do they know where to go? Bees cannot speak our words, and they do not send text messages. Instead, they tell one another with an old and beautiful system: **dancing on the honeycomb**.

② The Austrian scientist **Karl von Frisch** spent most of his life watching bees through glass windows of their hives. In 1973 he was given the **Nobel Prize** for his discovery: different dances carry different messages. The two most famous dances are the **round dance** and the **waggle dance**. Each one is a kind of small, clear "flying map" made by the bee's own body.

③ The **round dance** is easy. When a bee finds food that is **very close** to the hive, she runs in small circles, first to the right and then to the left, again and again. Her sisters follow her body closely, touching her with their antennae. The round dance says, more or less, "Good news — there is nectar **within about 50 metres**. Fly out and search!" It does **not** give a clear direction, because the food is so near that the bees can find it by smell and sight.

④ The **waggle dance** is more amazing. When the food is **far away**, the bee runs in a straight line while **shaking her belly** quickly — this is the "waggle". Then she curves back to the starting point, runs the straight line again, and keeps repeating. Two pieces of information are written into this dance. First, the **direction** of the straight line, compared with **straight up**, tells her sisters the angle between the food and the **sun**. Second, the **length** of the waggle run tells them **how far**: a longer waggle means a longer flight.

⑤ Think of a clock. If the bee waggles straight up on the honeycomb, the food is in the **direction of the sun**. If she waggles 60 degrees to the right of straight up, her sisters should fly 60 degrees to the right of the sun when they leave the hive. The longer she waggles — say, one second more — the farther away the food is. In this way, one small dance can easily talk about a flower field **three kilometres away**.

⑥ Scientists have also found that bees sometimes **change their dance** when they learn the hive is not safe. For example, if bees meet danger near a flower, they may do a short **"stop signal"** (a quick head-butt) to tell dancing sisters, "Please do not send more bees there!" This shows that bee communication is not just a fixed instinct, but a system that can **adjust to new information**. The next time we see a bee on a sunflower, we may

remember: behind every honey jar, there is a silent but rich language written in the steps of a dance.

根据文章内容，选择最佳答案。

1. Why does the writer describe bee behaviour as "an old and beautiful system"?

A. Because bees use modern phones.
B. Because bees dance on the honeycomb to share messages.
C. Because bees write small books.
D. Because bees speak many human languages.

2. What did Karl von Frisch win in 1973?

A. A football medal.
B. The Nobel Prize for discovering bee dances.
C. A singing prize.
D. The World Cup.

3. What does the round dance mainly tell other bees?

A. The food is very far away.
B. The food is dangerous.
C. The food is close, about within 50 metres.
D. The sun is gone.

4. In the waggle dance, what does the LENGTH of the straight run show?

A. The colour of the flower.
B. The number of enemies.
C. How far away the food is.
D. The weather.

5. What new finding does paragraph ⑥ share?

A. Bees can change their dance and even warn others about danger.
B. Bees always dance the same way.
C. Bees never see the sun.
D. Bees cannot find their way home.

答案与解析

1.
【答案】B
【题型】细节理解题

【定位】第①段末 "dancing on the honeycomb"
【解析】蜂在蜂巢上起舞。故选 B。

2.
【答案】B
【题型】细节理解题
【定位】第②段 "1973... Nobel Prize"
【解析】诺贝尔奖。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第③段 "within about 50 metres"
【解析】食物近，约50米内。故选 C。

4.
【答案】C
【题型】细节理解题
【定位】第④段 "the length... how far"
【解析】摇尾时长代表距离。故选 C。

5.
【答案】A
【题型】推理判断题
【定位】第⑥段 "stop signal"
【解析】蜜蜂会用"停止信号"更新信息。故选 A。

词汇注释（超纲词）

- **hive** [n.] 蜂箱
- **antenna** [n.] 触角（复数 antennae）
- **waggle** [v./n.] 摆动